

BCNF Analysis – Student, Course and Hobby

Problem

Consider the relation R(Student, Course, Hobby). The business rules state that a student's courses and hobbies are completely independent.

Student	Course	Hobby
Anu	Maths	Painting
Anu	Maths	Singing
Anu	Physics	Painting
Anu	Physics	Singing
Ravi	Chemistry	Dancing

Business Rules

- Student $\twoheadrightarrow$ Course
- Student $\twoheadrightarrow$ Hobby

These are multivalued dependencies because the set of courses and the set of hobbies are independent for a student.

BCNF Check

BCNF requires every non-trivial functional dependency $X \rightarrow Y$ to have X as a superkey. The stated rules are multivalued dependencies, not functional dependencies. Therefore, they do not by themselves create a BCNF violation.

4NF Observation

The more precise normal-form issue is Fourth Normal Form (4NF). The independent multivalued facts should be separated into two relations:

Student_Course(Student, Course)

Student_Hobby(Student, Hobby)

Conclusion

This example illustrates the distinction between BCNF and 4NF. BCNF addresses functional dependencies, while 4NF additionally addresses non-trivial multivalued dependencies. The Student–Course–Hobby example is therefore best described as a 4NF issue rather than a BCNF violation.